# MATEMATICI 2

## Curs 5

**Temă:**

Fie $V$ = sp. vect. finit peste $K$

și $\mathcal{B} = \{v_1, \dots, v_n\}$ = bază în $V$

Considerăm $K^n = \left\{ \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \alpha_i \in K \right\}$

cu legea int. $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 \\ \vdots \\ \alpha_n + \beta_n \end{pmatrix}$

Să se arate că $K^n$ e sp. vect. peste K cu legea int. $\alpha \cdot \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \vdots \\ \alpha a_n \end{pmatrix}$

$f_{\mathcal{B}} : V \to K^n$ ; $f_{\mathcal{B}}(v) = [v]_{\mathcal{B}}$ e izomorfism de sp. vect.

**T.:** privind dim nucleului și img. unei apl. liniare

Dc. $f : V \to W$ e apl. liniară cu $V, W$ = sp. vectoriale finite peste $K$.

atunci $\dim_K V = \dim_K \operatorname{Ker} f + \dim_K \operatorname{Im} f$

**Dem.:** Fie $f : V \to W$ liniară și $B_{\ker f} = \{v_1, \dots, v_p\} \subset V$ bază

Cum $B_{\ker f}$ = liniar indep. $\Rightarrow$ $B_{\ker f}$ se poate compl.

până la o bază a lui $V$ și avem:

$B_V = \{v_1, \dots, v_p, u_{p+1}, \dots, u_n\}$ ; $n = \dim_K V$

Vom arăta că $\{f(u_{p+1}), \ldots, f(u_n)\}$ = bază în $\mathrm{Im} f$

(lin. indep) Fie $\alpha_1 f(u_{p+1}) + \ldots + \alpha_{p+(n-p)} f(u_{p+(n-p)}) = 0$

$$f(\alpha_1 u_{p+1} + \ldots + \alpha_{p+(n-p)} u_{p+(n-p)}) = 0 \Rightarrow$$

$$\Rightarrow \alpha_1 u_{p+1} + \ldots + \alpha_{n-p} u_{p+(n-p)} \in \mathrm{Ker} f \Rightarrow$$

$\Leftrightarrow \exists\ \beta_1, \dots, \beta_p \in K$ a.î.

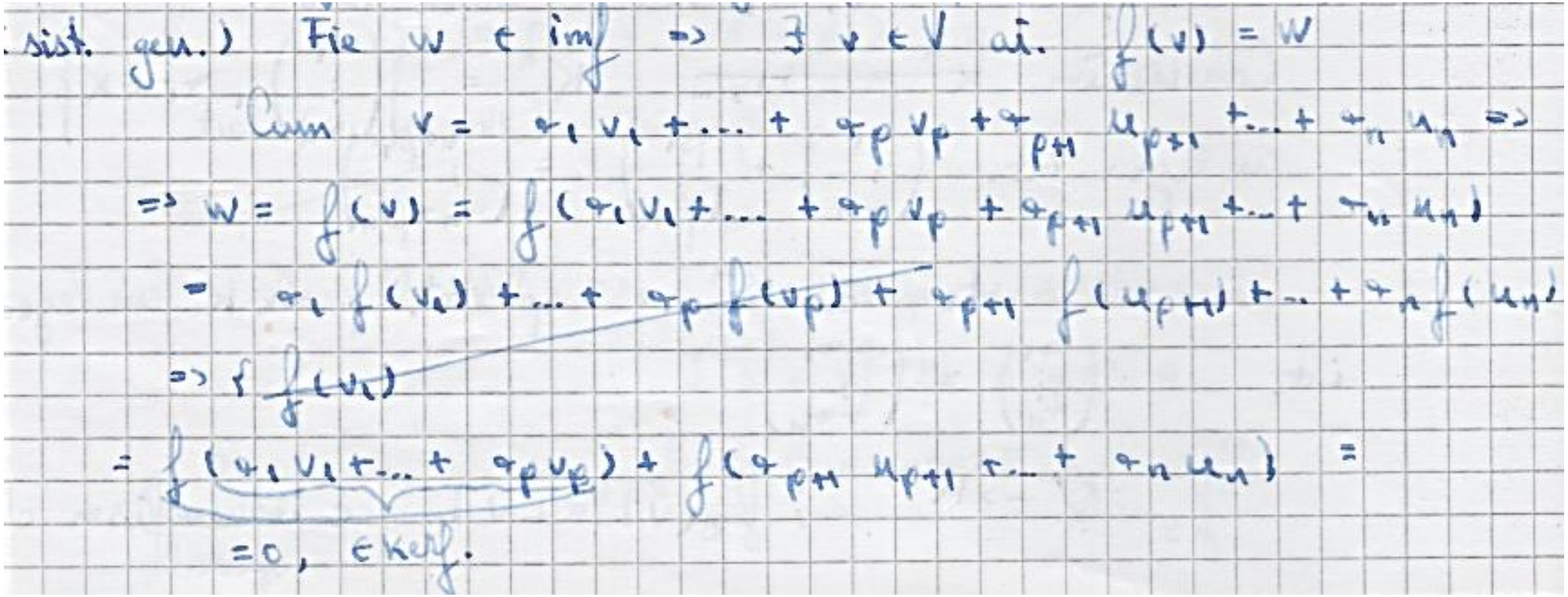
$$\alpha_1 u_{p+1} + \dots + \alpha_{n-p}\, u_{p+(n-p)} =$$

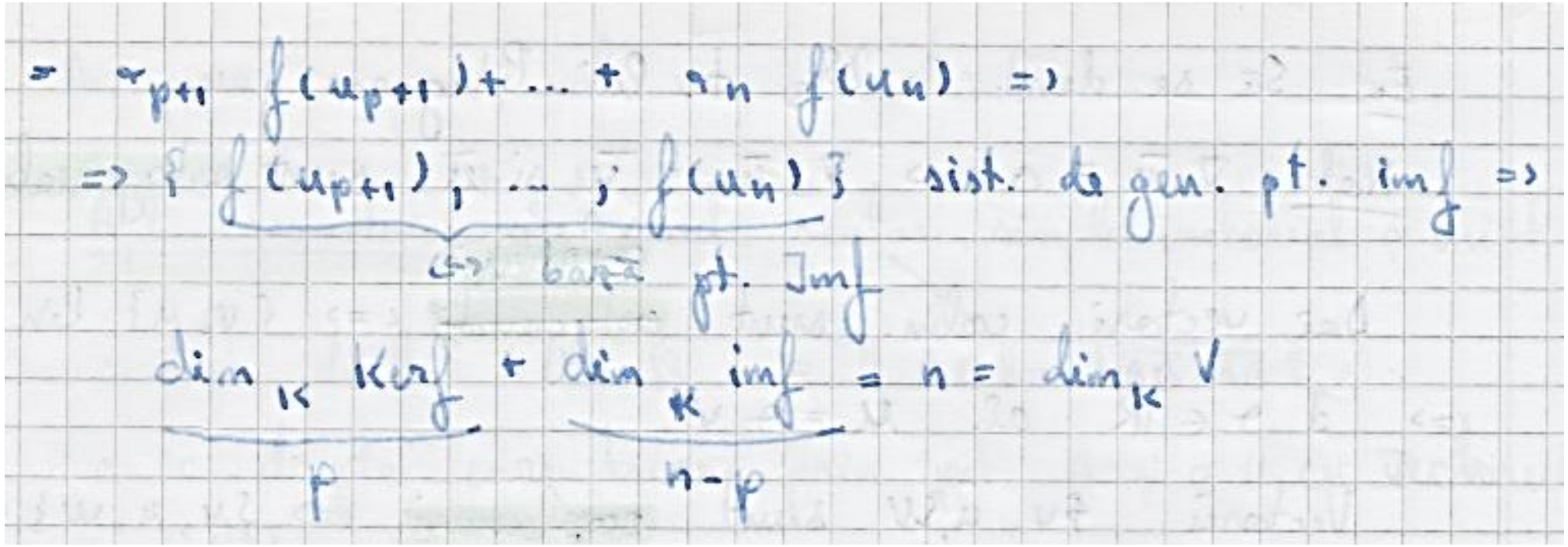
$$= \beta_1 v_1 + \dots + \beta_p v_p$$

$$(-\beta_1)v_1 + \ldots + (-\beta_p)v_p + \gamma_1 u_{p+1} + \ldots + \gamma_{n-p} u_{p+(n-p)} = 0$$

$$\Rightarrow \beta_1 = \ldots = \beta_p = \gamma_1 = \ldots = \gamma_{n-p} = 0 \Rightarrow$$

$$\Rightarrow \{f(u_{p+1}), \ldots, f(u_{p+(n-p)})\} \text{ lin. indep.}$$


sist. gen.) Fie $w \in \mathrm{im} f \Rightarrow \exists\, v \in V$ a.i. $f(v) = w$
Cum $v = a_1 v_1 + \dots + a_p v_p + a_{p+1} u_{p+1} + \dots + a_n u_n \Rightarrow$
$\Rightarrow w = f(v) = f(a_1 v_1 + \dots + a_p v_p + a_{p+1} u_{p+1} + \dots + a_n u_n)$
$= a_1 f(v_1) + \dots + a_p f(v_p) + a_{p+1} f(u_{p+1}) + \dots + a_n f(u_n)$
$\Rightarrow$ f
$= f(a_1 v_1 + \dots + a_p v_p) + f(a_{p+1} u_{p+1} + \dots + a_n u_n) =$
$= 0$, $\in \ker f$.


= α_{p+1} f(u_{p+1}) + ... + α_n f(u_n) =>
=> {f(u_{p+1}), ..., f(u_n)} sist. de gen. pt. Imf =>
<=> bază pt. Imf
dim_K Kerf + dim_K Imf = n = dim_K V
p
n-p

Fie $M = \mathbb{R}^3 \times \mathbb{R}^3 = \{ (A,B) \mid A,B \in \mathbb{R}^3 \}$

Fie $\rho$ = rel. binară pe M def. prin $(A_1,B_1) \rho (A_2,B_2) \Leftrightarrow$ mijl. lui $A_1B_2$ coincide cu mijl. lui $A_2B_1$.

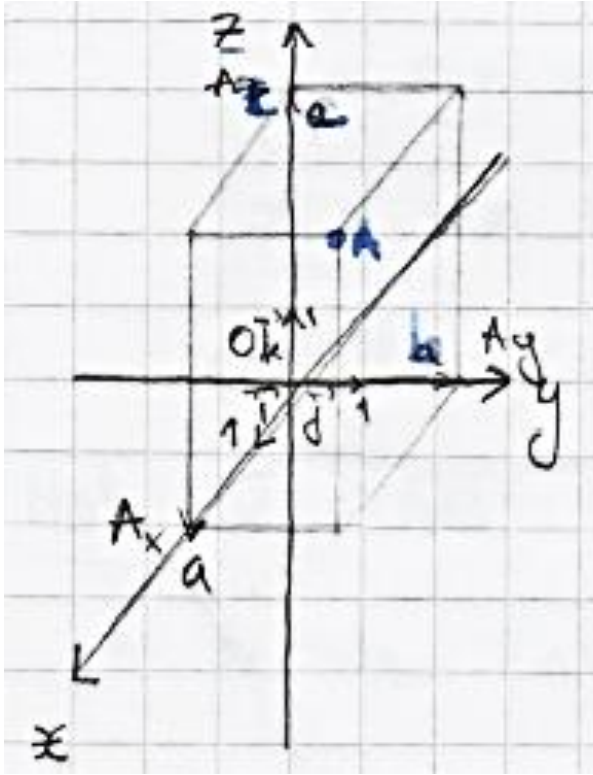
$\Rightarrow \rho$ = rel. binară de echiv.

$M/_\rho = \{ (A,B) \overset{\text{notăm}}{=} \widehat{AB} \mid (A,B) \in M \}$

Not $M/_S = V_3$ (vectorii din spațiu)

Def. înm. cu un scalar (= nr real) a unui elem. din $V_3$ similar celui (de la licee) de la înm. unui nr. cu un vector plan și, corn., suma elem. din $V_3$ conf. regulii paralelogramului sau regulii $\Delta \Rightarrow$

$\Rightarrow V_3$ devine sp. vectorial real.




$$\overline{OA} = \overline{OA_x} + \overline{OA} + \overline{OA_z}$$

$$= a\vec{i} + b\vec{j} + c\vec{k}$$

# Produs scalar

$\bar{v} = a\bar{i} + b\bar{j} + c\bar{k}$

$\bar{u} = l\bar{i} + m\bar{j} + n\bar{k}$

$\bar{v}\bar{u} = al + bm + cn$

$\|\bar{v}\| = \sqrt{\bar{v}\cdot\bar{v}} = \sqrt{a^2+b^2+c^2}$ = lungimea lui $\bar{v}$ (

$\cos(\bar{v},\bar{u}) = \dfrac{\bar{v}\cdot\bar{u}}{\|\bar{v}\|\cdot\|\bar{u}\|}$

**Ex.:** Să se dem. în $\mathcal{V}_3$ t. lui Pitagora gen.

Not. $\bar{v} \cdot \bar{u} = 0 \iff \bar{v} \perp \bar{u}$ ; $\bar{v}$ și $\bar{u}$ sunt ortogonali.

Doi vectori ~~colin.~~ coliniari sunt ortogonali $\iff \{v, u\}$ lin. indep. $\iff$

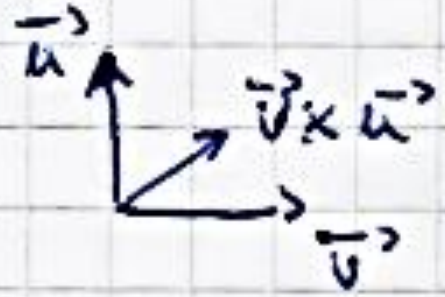
$\iff \exists\, \alpha \in \mathbb{R}$ a.î. $u = \alpha v$

Vectorii $\{v, u, w$ sunt coplanari $\iff \{v, u, w\}$ lin. indep $\iff$

$\iff \exists\, \alpha, \beta \in \mathbb{R}$ a.î. $v = \alpha u + \beta w$

# Produs vectorial

$\bar{v} \times \bar{u} = \bar{w} \Leftrightarrow \quad \bar{w} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a & b & c \\ l & m & n \end{vmatrix}$



**Prop.**

1) $\vec{v} \times \vec{u} = -\vec{u} \times \vec{v}$

2) $\vec{v} \times (\alpha \vec{u}_1 + \beta \vec{u}_2) = \alpha \vec{v} \times \vec{u}_1 + \beta \vec{v} \times \vec{u}_2$

3) $\|\vec{v} \times \vec{u}\| = \|\vec{v}\| \, \|\vec{u}\| \sin(\vec{v}, \vec{u})$

(Aria paralelog. în care $v$ și $u$ sunt laturi)

# Produs mixt vectorial

Dc. $\bar{v}_1 = l_1 \bar{i} + m_1 \bar{j} + n_1 \bar{k}$

$\bar{v}_2 = l_2 \bar{i} + m_2 \bar{j} + n_2 \bar{k}$

$\bar{v}_3 = l_3 \bar{i} + m_3 \bar{j} + n_3 \bar{k}$

$$\bar{v}_1 \cdot (\bar{v}_2 \times \bar{v}_3) = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$$

$\| \text{not.}$

$(\bar{v}_1, \bar{v}_2, \bar{v}_3)$

**Prop.** 1) $\bar{v}_1 \cdot (\bar{v}_2 \times \bar{v}_3)$ = vol. paralelipipedului

2) liniaritate în rap. cu fiecare argument.

3) (Dc. se inv. 2 vectori vecini, se schimbă semnul prod.)

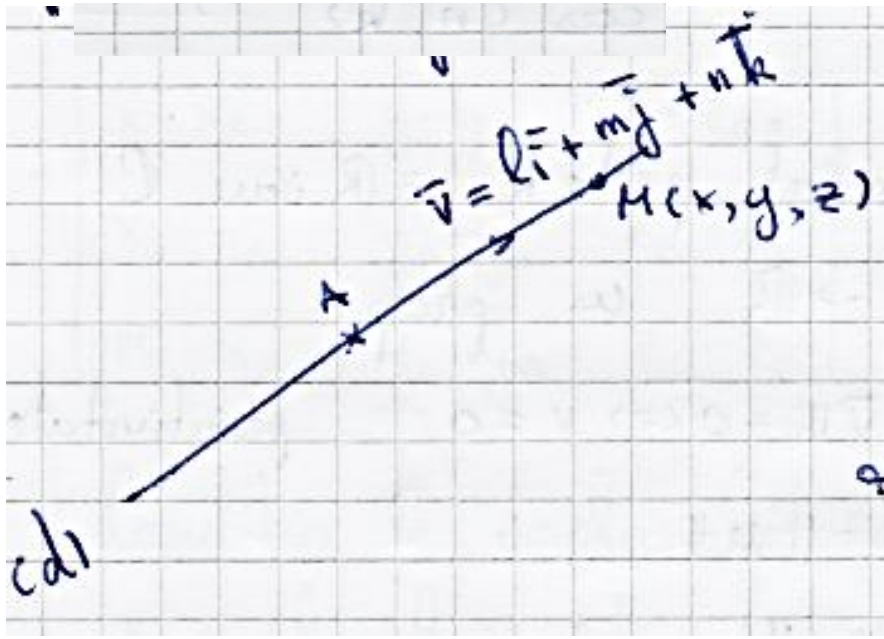
Dc. se schimbă ord. prod. mixt printr-o perm. impară, prod. mixt își schimbă semnul.

Dc. perm. e pară, val. nu se schimbă.

# Aplicații geometrice

ale prod. scalar, simplu vectorial și mixt.

- dist. de la A la B $= \|\overline{AB}\| = \sqrt{\overline{AB}\cdot\overline{AB}}$ ℰ
- ec. dreptei care trece prin pct. A și e ∥ cu vectorul $\bar{v} = l\bar{i} + m\bar{j} + n\bar{k}$



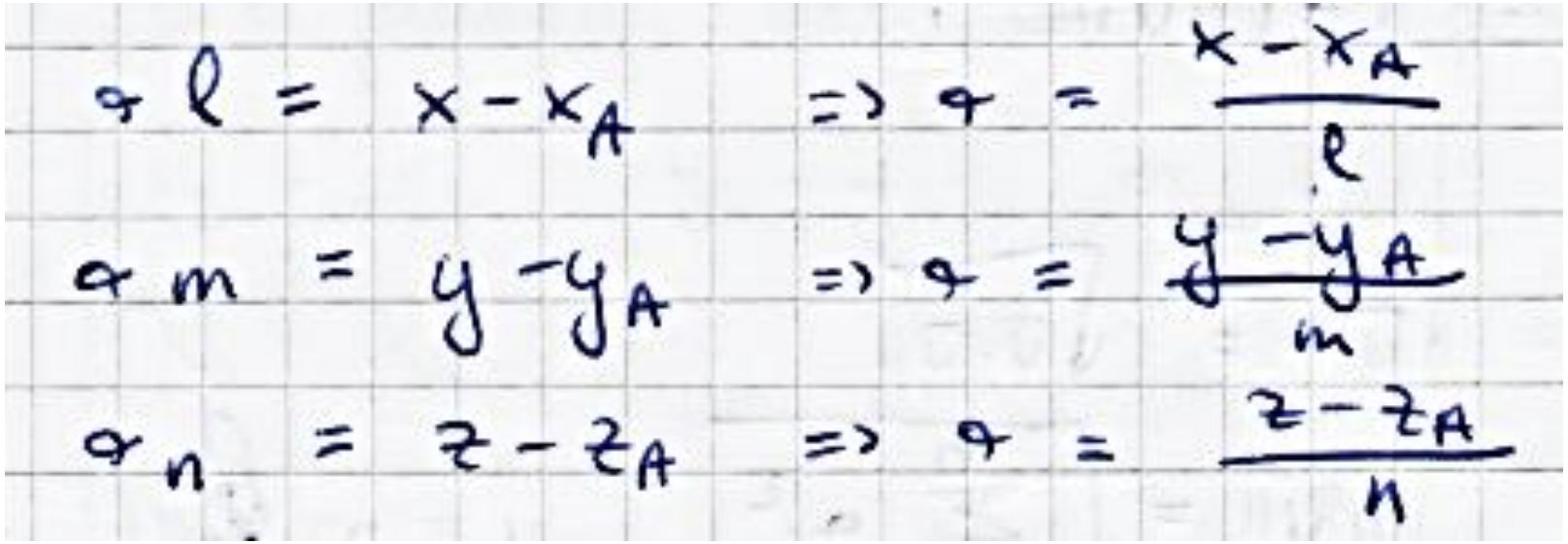

$M \in d \Leftrightarrow \overline{AM}$ coliniar cu $\bar{v}$

$(\overline{AM} \parallel \bar{v})$

$\Leftrightarrow \exists\, \alpha \in \mathbb{R}$ a.î. $\alpha \bar{v} = \overline{AM}$

$\alpha(l\bar{i} + m\bar{j} + n\bar{k}) = (x - x_A)\,i + (y - y_A)\,j +$

$+ (z - z_A)\,k$


$q\,l = x - x_A \Rightarrow q = \frac{x - x_A}{l}$
$q\,m = y - y_A \Rightarrow q = \frac{y - y_A}{m}$
$q\,n = z - z_A \Rightarrow q = \frac{z - z_A}{n}$

$$\frac{x - x_A}{l} = \frac{y - y_A}{m} = \frac{z - z_A}{n}$$ = ec. ~~distanței~~ dr. (d) sub forma canonică

$x = \alpha l + x_A$

$y = \alpha m + y_A$ , $\alpha \in \mathbb{R}$ ec. parametrică a lui (d)

$z = \alpha n + z_A$

• ec. dr. $(d)$ care trece prin $A$ și $B$

Not. $\bar{v} = \overline{AB} = (x_B - x_A)\,\bar{i} + (y_B - y_A)\,\bar{j} + (z_B - z_A)\,\bar{k}$

ec. dr. care trece prin $A$ și e $\parallel$ cu $\bar{v}$ este:

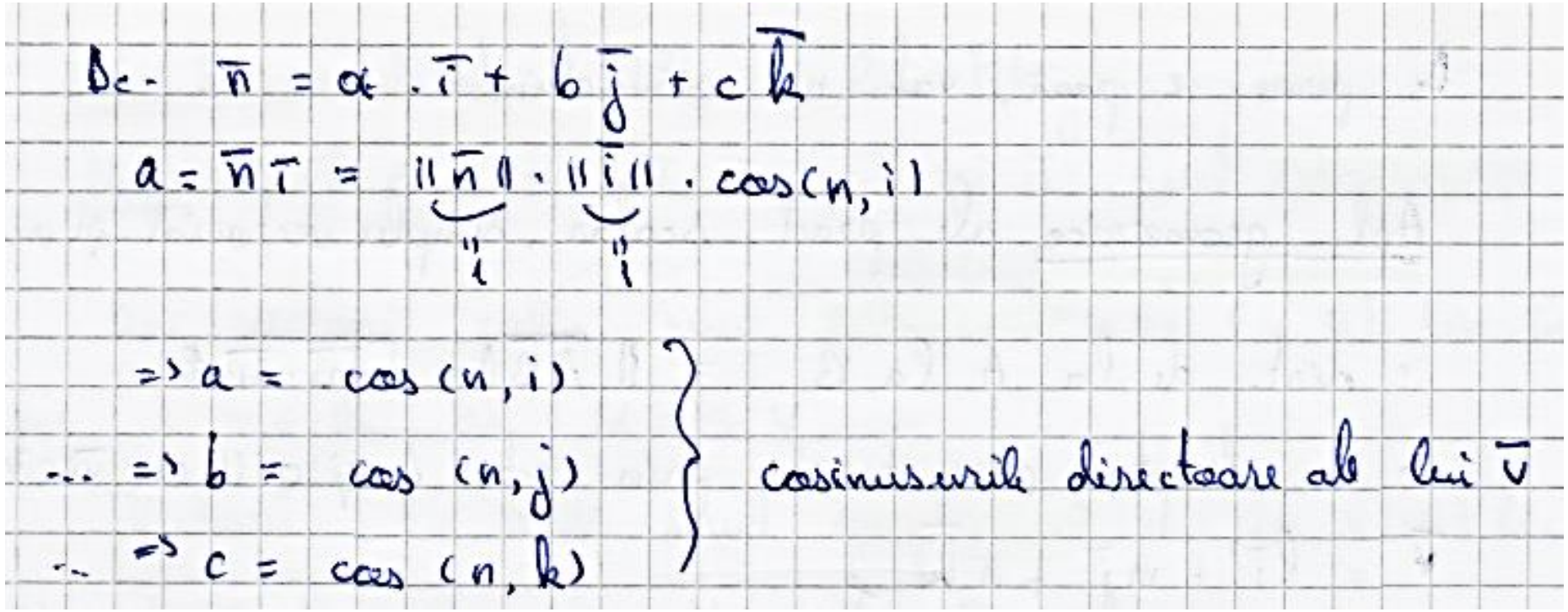
$$\frac{x - x_A}{x_B - x_A} = \frac{y - y_A}{y_B - y_A} = \frac{z - z_A}{z_B - z_A}$$

**Def.** Dc. $\|\bar{n}\| = 1$ și $\bar{v} = a\bar{n}$, $a > 0$ se spune că $\bar{n}$ este versorul lui $\bar{v}$.

Din rel. $\bar{v} = a\bar{n}$, $\|\bar{n}\| = 1$

$$\|\bar{v}\| = \|a \cdot \bar{n}\| = a \cdot \|\bar{n}\| \Rightarrow \|\bar{v}\| = \|\bar{n}\| \cdot \underbrace{|a|}_{>0} \Rightarrow$$

$$\Rightarrow a = \|\bar{v}\| \Rightarrow \|\bar{n}\| = \frac{1}{\|\bar{v}\|} \quad n = \frac{1}{\|\bar{v}\|} \cdot \bar{v}$$


Dc. $\bar{n} = a \cdot \bar{i} + b\bar{j} + c\bar{k}$
$a = \bar{n}\bar{i} = \|\bar{n}\| \cdot \|\bar{i}\| \cdot \cos(n, i)$
$\|\bar{n}\| = 1$
$\|\bar{i}\| = 1$
$\Rightarrow a = \cos(n, i)$
$\Rightarrow b = \cos(n, j)$
$\Rightarrow c = \cos(n, k)$
cosinusurile directoare ale lui $\bar{v}$

**Def.**

Fie $V$ = sp. vectorial peste $K = \mathbb{R}$ sau $\mathbb{C}$

Aplicatia $\|\cdot\| : V \to \mathbb{R}$ cu prop.

1) $\|\cdot\| \geq 0$, $\|\vec{v}\| = 0 \Leftrightarrow v = 0$ pozitivitate

2) $\|\vec{v} + \vec{u}\| \leq \|v\| + \|u\|$

3) $\|a v\| = |a| \|v\|$,

sn. norma.

**Ex.** $V = \mathcal{V}_3$, $\|\vec{v}\| = \sqrt{\vec{v}\cdot\vec{v}}$

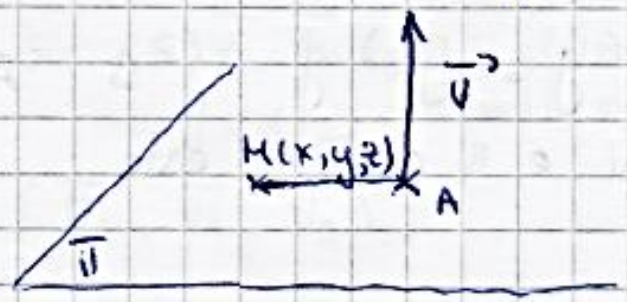
1) $V = \mathbb{R}^n$, $\|v\| = \sqrt{\sum_{i=1}^{n} a_i^2}$, $v = a_1 e_1 + \dots + a_n e_n$,

$e_k = (0, 0, \dots, 0, \underset{\text{pe } k}{1}, 0, \dots, 0)$

• ec. planului det. cu un pct. din plan ($A$) și un vector ($\bar{v}$) $\perp$ pe plan

Fie $A(x_A, y_A, z_A) \in (\pi)$ = plan

$\bar{v} = l\bar{i} + m\bar{j} + n\bar{k}$




ec. vectorială a planului

$M \in \pi \Leftrightarrow \overline{AM} \perp \bar{v} \Leftrightarrow \overline{AM} \cdot \bar{v} = 0$

// ($a\bar{v} = \overline{AM}$) ec. vectorială a dr.

$$\left[(x-x_A)\bar{i}+(y-y_A)\bar{j}+(z-z_A)\bar{k}\right]\cdot(l\bar{i}+m\bar{j}+n\bar{k})=0$$

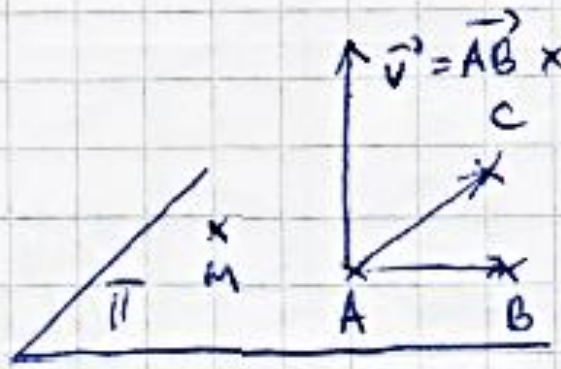
$$(x-x_A)\,l+(y-y_A)\,m+(z-z_A)\,n=0$$

$$x\,l+y\,m+z\,n-D=0 \quad \text{ec. gen. a planului}$$

$$D=x_A l+y_A m+z_A n$$

• ec. planului care trece prin 3 puncte

Fie $A, B, C \in \pi$ , $\pi = ?$




$M(x, y, z)$

$M \in \pi \Leftrightarrow A\,M \perp v \Leftrightarrow$

$\Leftrightarrow \overrightarrow{AM} \cdot v = 0 \Rightarrow$

$\Rightarrow \overrightarrow{AM} \cdot (\overrightarrow{AB} \times \overrightarrow{AC}) = 0$

$$\begin{vmatrix} x - x_A & y - y_A & z - z_A \\ x_B - x_A & y_B - y_A & z_B - z_A \\ x_C - x_A & y_C - y_A & z_C - z_A \end{vmatrix} = 0$$

! Condiția de coplanaritate a pctelor. $A, B, C, D \Leftrightarrow$

$\Leftrightarrow D \in$ planului det. de $A, B$ și $C$

$$\begin{vmatrix} x_D - x_A & y_D - y_A & z_D - z_A \\ x_B - x_A & y_B - y_A & z_B - z_A \\ x_C - x_A & y_C - y_A & z_C - z_A \end{vmatrix} = 0$$

$\alpha_i \in \mathbb{R}$

**Ex.** Să se afle dist. de la pct. C la dr. care trece prin A și B.

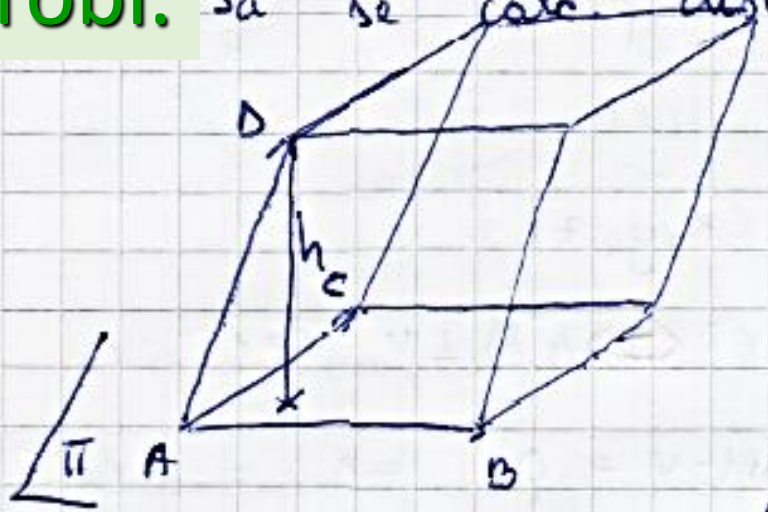
$$d_{AB}: \quad \frac{x-x_A}{x_B-x_A} = \frac{y-y_A}{y_B-y_A} = \frac{z-z_A}{z_B-z_A}$$



$$\text{dist}(C, AB) = \text{Aria}_{\square ABC} \cdot \frac{1}{\text{lungimea } AB} =$$

$$= \frac{|AB \times AC|}{|\overline{AB}|}$$

**Probl.** Să se calc. dist. de la D la planul (A, B, C).




$$\text{dist}(D, \pi(ABC)) = \frac{\text{vol paralelipiped}}{A_{\text{bază}}} =$$

$$= \frac{|\overline{AD} \cdot (\overline{AB} \times \overline{AC})|}{\| \overline{AB} \times \overline{AC} \|}$$

Volum tetraedru $(ABCD) = \frac{1}{6} \left| \overline{AD} \cdot (\overline{AB} \times \overline{AC}) \right|$

probl. Să se det. dist. dintre dreptele $D_1$ și $D_2$ din spațiu

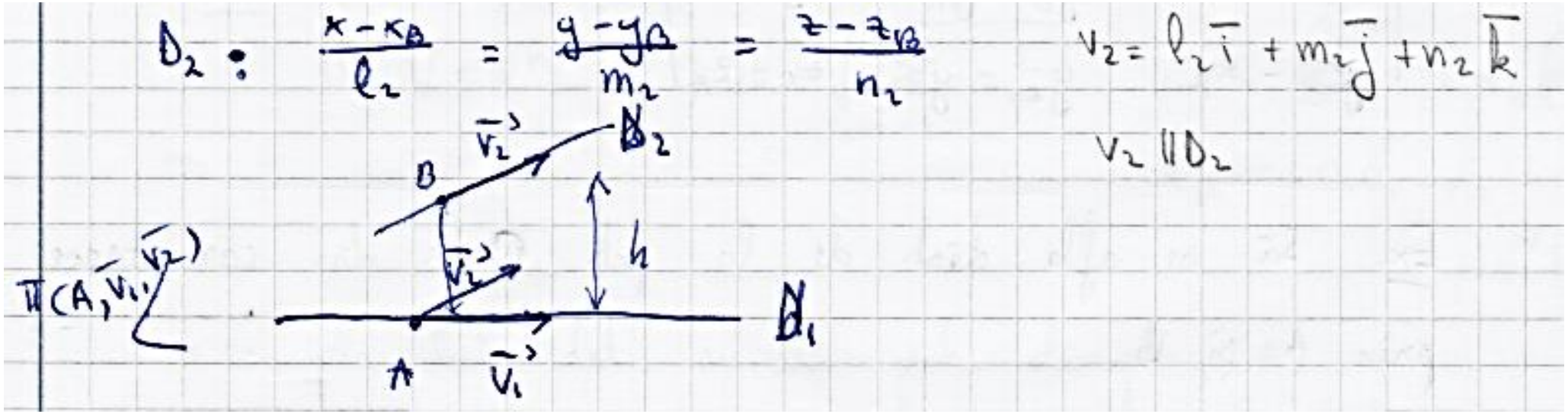
$$D_1: \frac{x - x_A}{l_1} = \frac{y - y_A}{m_1} = \frac{z - z_A}{n_1}$$

$$\overline{v_1} = l_1 \overline{i} + m_1 \overline{j} + n_1 \overline{k} \parallel D_1$$

$D_2 : \dfrac{x-x_B}{l_2} = \dfrac{y-y_B}{m_2} = \dfrac{z-z_B}{n_2}$

$v_2 = l_2\bar{i} + m_2\bar{j} + n_2\bar{k}$

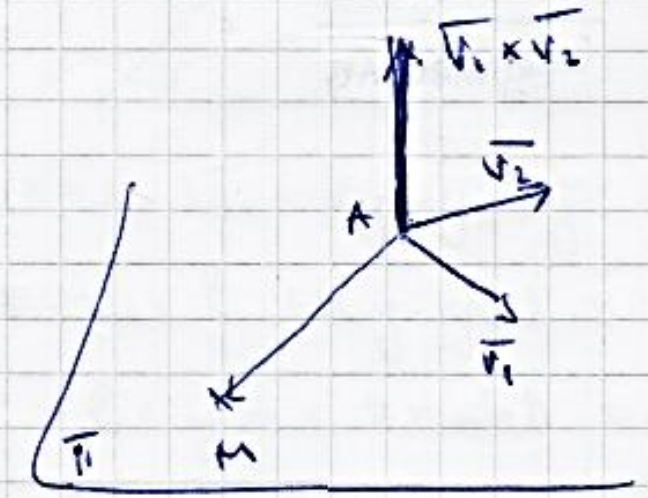
$v_2 \parallel D_2$



$$h = dist(D_2, D_1) = dist(D_2, \Pi(A, \bar{v}_1, \bar{v}_2))$$

$$h = \frac{\text{Vol. paralelipiped}(\bar{v}_1, \bar{v}_2, \overline{AB})}{\text{Arie bază } (v_1, v_2)} = \frac{|\bar{v}_1 \cdot (\bar{v}_2 \times \overline{AB})|}{\|\bar{v}_1 \times \bar{v}_2\|}$$

$A \in \pi$ , $\bar{v}_1, \bar{v}_2 \parallel \pi$



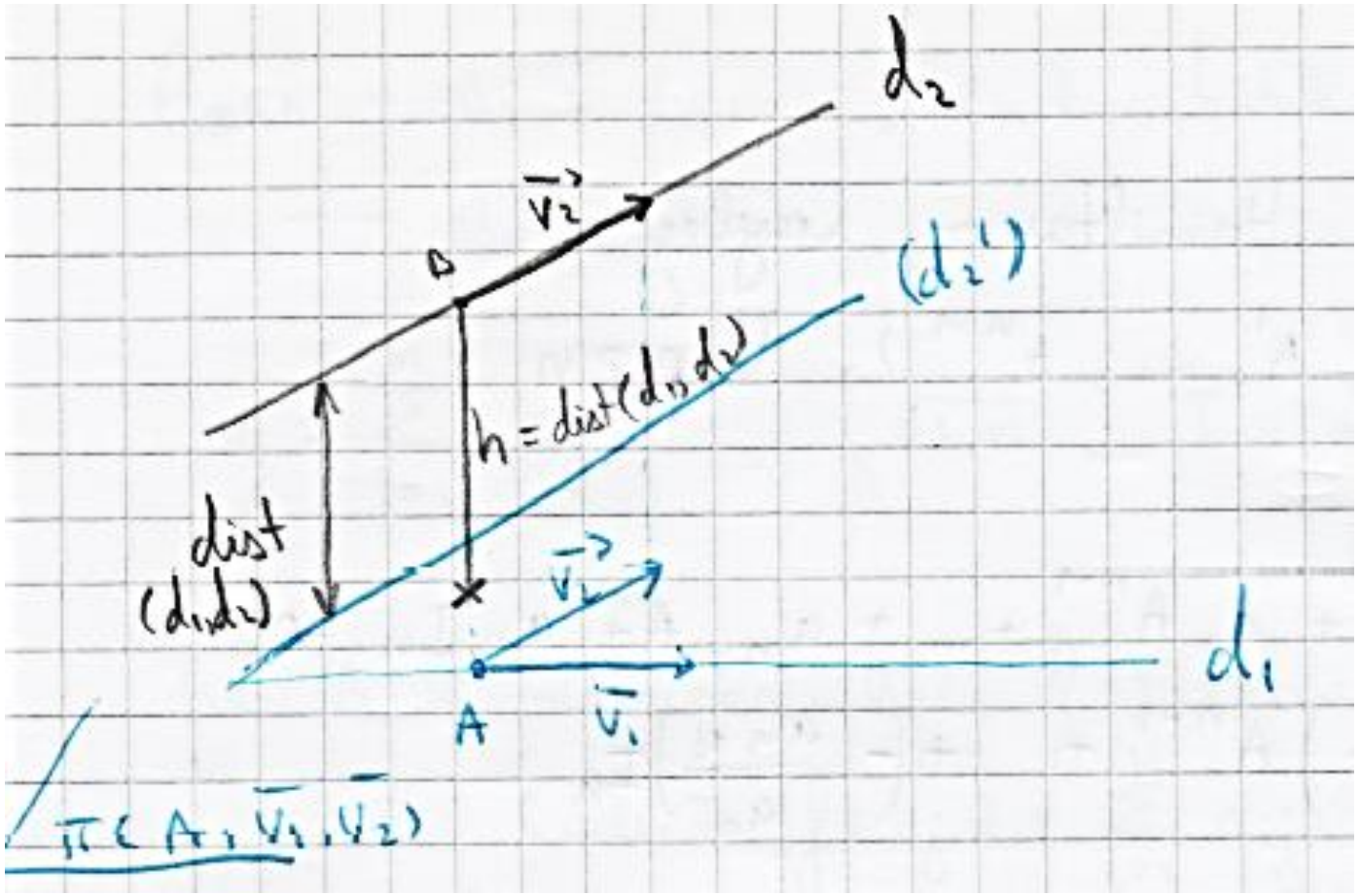

$M \in \pi \Leftrightarrow \overline{AM} \perp \bar{v}_1 \times \bar{v}_2 \Leftrightarrow$

$\Leftrightarrow \overline{AM} \cdot (\bar{v}_1 \times \bar{v}_2) = 0 \Leftrightarrow$

$$\begin{vmatrix} x - x_A & y - y_A & z - z_A \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

$\bar{v}_1 = l_1 \bar{i} + m_1 \bar{j} + n_1 \bar{k}$

$\bar{v}_2 = l_2 \bar{i} + m_2 \bar{j} + n_2 \bar{k}$


$d_2$
$\vec{v_2}$
B
$(d_2')$
$h = dist(d_1, d_2)$
dist
$(d_1, d_2)$
$\vec{v_2}$
$d_1$
A
$\vec{v_1}$
$\pi(A, \vec{v_1}, \vec{v_2})$

**Temă:** 1) Să se det. sim. pct. $P(1,2,3)$ în rap. cu dr.

$d: \frac{x-1}{7} = \frac{y+1}{5} = \frac{z}{2}$

2) Să se det. sim. pct. $P(1,2,3)$ față de planul cu ec. gen. $2x - 3y + z + 7 = 0$